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predators of lac insects. The caterpillars of these parasites feed upon lac insects showing hyper-parasitism.

Hyper-parasitism - A condition in which a secondary parasite develops within a previously existing parasite or a hyperparasite is the parasite whose host is also a parasite.

Lac cultivation is a complicated process, so the cultivators should know well about the inoculation, swarming period and harvesting of lac. The process of introducing lac insect on the host plant is called inoculation. Before inoculation, pruning of the host plant is done. The twigs having brood lac, i.e., lac insect about 20 cm in length are attached to fresh host plants. The lac insect then repeats its life cycle. The collection of lac from the host plant is known as harvesting. Harvesting may be done before swarming (immature) or after swarming (mature). Immature harvesting produces 'Ari lac' whereas mature harvesting produces the mature lac. Lac cut from the host plant is called 'Stick lac'. The lac present on the twig is scraped and collected. After grinding, the unnecessary materials like dusts and fine particles are removed. The resultant lac is called 'seed lac'. The seed lac is sun dried and then melted to produce 'shellac'.

Economic importance of Lac

a. Lac is largely used as a sealing wax and adhesive for optical instruments. It is used in electric industry, as it is a good insulator.
b. It is used in preparations of shoe and leather polishes and as a protective coating of wood.
c. It is used in laminating paper board, photographs, engraved materials and plastic moulded articles.
d. Used as a filling material for gold ornaments

13.6 Aquaponics

Aquaponics is a technique which is a combination of aquaculture (growing fish) and hydroponics (growing plants in non-soil media and nutrient-laden water). Aquaponics may also prevent toxic water runoff. It also maintains ecosystem balance by recycling the waste and excretory products produced by the fish. In India, aquaponics was started in 2013. Some primary methods of aquaponic gardening that are in use nowadays are as follows: (i) **Deep water culture** is otherwise known as raft based method. In this method a raft floats in water. Plants are kept in the holes of raft and the roots float in water. This method is applicable for larger commercial scale system. By this method fast growing plants are cultivated. (ii) **Media based method** involves growing plants in inert planting media like clay pellets or shales. This method is applicable for home and hobby scale system. Larger number of fruiting plants, leafy green plants, herbs and other varieties of plants can be cultivated (Figure 13. 8) (iii) **Nutrient Film technique** involves the passage of nutrient rich water through a narrow trough or PVC pipe. Plants are kept in the holes of the pipe to allow the roots to be in free contact with in the water stream. (iv) **Aqua vertica** is otherwise known as vertical aquaponics. Plants are stacked on